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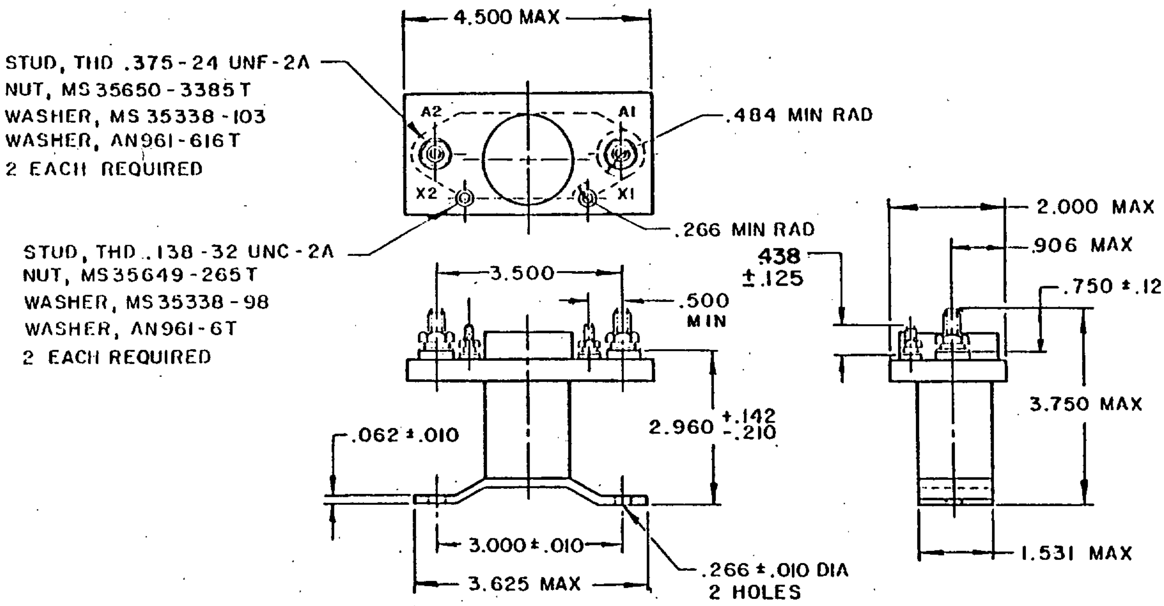

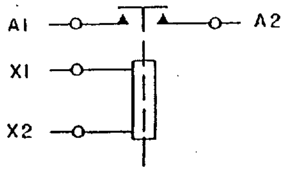

CIRCUIT (applicable to configurations A and B)

CONFIGURATION A. Outline drawing (for details see tables I and II).

Ⓗ Denotes Changes

PREPARING ACTIVITY: AIR FORCE-85 CUSTODIANS: ARMY- NAVY- AS AIR FORCE-85 DLA- REVIEW: NAVY-AS, EC AIR FORCE-11, 99 USER: PROJECT NUMBER: 5945-0841-01	MILITARY SPECIFICATION SHEET TITLE RELAYS, ELECTROMAGNETIC, 200 AMPERES, 1PST (N O), TYPE III, NON-HERMETICALLY SEALED	SPECIFICATION SHEET NUMBER MS24172H 25 July 1990 SUPERSEDING MS24172G 24 Jan 1990 AMSC N/A FSC 5945
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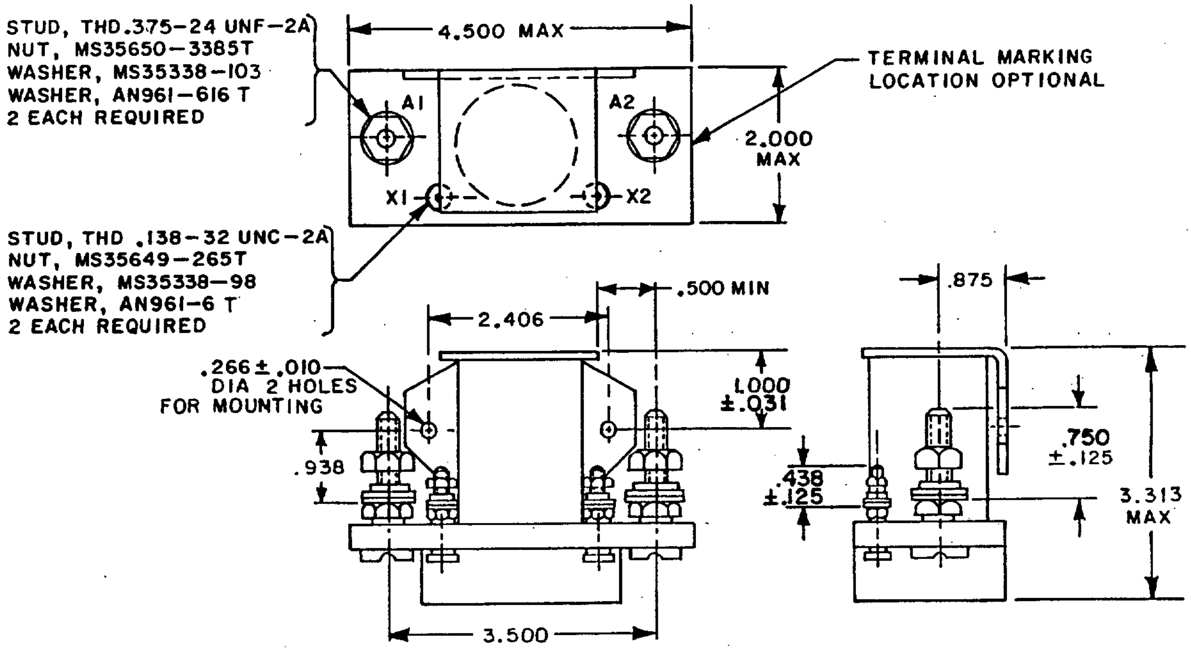

CONFIGURATION B. Outline drawing (for details see tables I and II).

Inches	mm	Inches	mm	Inches	mm
.010	0.25	.484	12.30	2.000	50.80
.062	1.57	.500	12.70	2.406	61.11
.125	3.18	.750	19.06	3.960	75.18
.138	3.51	.813	20.66	3.000	76.20
.142	3.61	.875	22.22	3.313	84.15
.210	5.33	.906	23.01	3.500	88.90
.266	6.76	.938	23.83	3.750	95.26
.375	9.52	1.000	25.40	4.500	114.30
.438	11.13	1.531	38.89		

NOTES:

1. Dimensions are in inches.
2. Terminal cover not required.
3. Additional flat washer may be used for terminal seat.
4. Part number MS24172-D1 replaces part no. MS24172-1.
5. This specification sheet takes precedence over documents referenced herein.
6. Referenced documents shall be of the issue in effect on date of invitation for bid.
7. Metric equivalents are given for general information only.
8. Unless otherwise specified, tolerances are ±.062 (1.57 mm).
9. Terminal temperature rise under continuous current conditions, 95°C. Intermediate current to be conducted at 71°C.
10. Shape of relay optional within envelope dimensions.

PREPARING ACTIVITY: AF-85
CUSTODIANS: ARMY - NAVY - AS
AIR FORCE - 85 DLA -
REVIEW: NAVY-AS, EC
AF-11, 99
USER:
PROJECT NUMBER: 5945-0841-01

MILITARY SPECIFICATION SHEET
TITLE
RELAYS, ELECTROMAGNETIC,
200 AMPERES, 1PST (N O),
TYPE III, NON-HERMETICALLY
SEALED

SPECIFICATION SHEET NUMBER
MS24172H
SUPERSEDING
MS24172G 24 Jan 1990
AMSC N/A FSC 5945

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TABLE I. Dash numbers and characteristics.

Dash number MS24172-	Type	Coil	Terminal type	Mounting or mating socket	Max weight in pounds
D1 1/	III	dc	Stud	Bracket - bottom	1.33
D2 2/	III	dc	Stud	Bracket - side	1.33

1/ For Government logistics support MS24172-D1 shall be used in lieu of AN3371-2.

2/ For Government logistics support MS24172-D2 shall be used in lieu of AN3371-1.

TABLE II. Operating characteristics.

MS part no.	Coil data											Time (milliseconds - maximum)					
MS24172	Coil	Nominal			Max		Max pick-up voltage			Hold volt-age 2/	Drop out voltage 2/	Operate 3/	Release 4/	Bounce			
														Main		Aux	
		Volts 1/	Freq Hz	ResΩ minimum 25°C +15% -10%	Volts	Amperes	Normal 2/	High temp test	Cont current test					NO	NC	NO	NC
D1	X1, X2	28	dc	10	29	4.5	7.5	11.5	---	3.0	.5	25	18	5.0			
D2	X1, X2	28	dc	10	29	4.5	7.5	11.5	---	3.0	.5	25	18	5.0			

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

PREPARING ACTIVITY: AF-85 **CUSTODIANS:** ARMY - NAVY - AS AIR FORCE - 85 DLA - **REVIEW:** NAVY-AS,EC AF-11,99 **USER:** **PROJECT NUMBER:** 5945-0841-01	**MILITARY SPECIFICATION SHEET** **TITLE** RELAYS, ELECTROMAGNETIC, 200 AMPERES, 1PST (N O), TYPE III, NON-HERMETICALLY SEALED	**SPECIFICATION SHEET NUMBER** MS24172H **SUPERSEDING** MS24172G 24 Jan 1990 **AMSC** N/A **FSC** 5945
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Ⓗ TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase 1/				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 HZ	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	50	200												
Inductive	10	100												
Inductive														
Motor	50	200												
Lamp														
Transfer load														2/
Mechanical life reduced current	100	50												
Intermediate current														

1/ Absence of value indicates relay is not rated for 3 phase application.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating indicated.

PREPARING ACTIVITY: AF-85
CUSTODIANS: ARMY - NAVY - AS
AIR FORCE - 85 DLA -
REVIEW: AF-11,99
NAVY AS, EC
USER:
PROJECT NUMBER: 5945-084[illegible]-01

MILITARY SPECIFICATION SHEET

TITLE
RELAYS, ELECTROMAGNETIC,
200 AMPERES, 1PST (N O),
TYPE III, NON-HERMETICALLY
SEALED

SPECIFICATION SHEET NUMBER
MS24172H
SUPERSEDING
MS24172G 24 Jan 1990
AMSC N/A FSC 5945

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ENVIRONMENTAL CHARACTERISTICS

Temperature range	-55°C to +71°C
Maximum altitude rating	50,000 ft
Shock G-level	25 g's
Duration	6-9 ms
Max duration contact opening	2 ms
Vibration - Sinusoidal (see table IV)	
Vibration - Random	N/A
High shock	N/A
Acceleration	10 g's

TABLE IV. Vibration levels.

5-10 Hz	10-55 Hz	55-250 Hz	250-500 Hz	500-1,500 Hz
.08 DA	.06 DA	2 g's	2 g's	

Qualification by similarity (see table V)

TABLE V. Qualification by similarity.

Part number MS24172-	Loads						Dynamics 1/			Environmental		
	Type 1			Type 1 ER								
	A	B	C	D	E	F	A	B	C	A	B	C
D1	4						2			4		
D2	4						2			4		

1/ For Group C testing, each dash number shall be tested to meet vibration requirements.

ELECTRICAL CHARACTERISTICS

Insulation resistance initial	100 megohms
After life or environmental tests	50 megohms
Dielectric strength (sea level)	2-5 seconds

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case	1,250 V rms	N/A	1,000 V rms	N/A
Aux contacts	1,250 V rms	N/A	1,000 V rms	N/A
All other points	1,250 V rms	N/A	1,000 V rms	N/A

Dielectric strength (altitude) 1 minute

	28 V dc	115 V ac
Coil to case	500 V rms	N/A
Aux contacts	500 V rms	N/A
All other points	700 V rms	N/A

Max contact drop initial	0.150 volt
After life test	0.175 volt
Overload current	(NO) 1,600 amperes
Rupture current	(NO) 2,000 amperes
Duty rating	Intermittant
RFI spec	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

PREPARING ACTIVITY: AF-85
CUSTODIANS: ARMY - NAVY - AS
AIR FORCE - 85 DLA -
REVIEW: AF-11,99
NAVY-AS,EC
USER:
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MILITARY SPECIFICATION SHEET

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200 AMPERES, 1PST (N O),
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